Pouya Mansournia
Senior Robotics Systems Engineer

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Portfolio | Türkiye | Open to Relocation

SUMMARY

Senior Robotics Systems Engineer with 8+ years of experience leading large-scale robotics, warehouse automation, and autonomous systems across logistics, industrial, and biomedical domains. Delivered production systems supporting 600K+ daily throughput and ultra-precision robotic platforms with international deployment.
Specialized in robotics architecture, embedded systems, and end-to-end autonomous system development from prototype to production. Co-founder-level ownership across the full product lifecycle, including hardware, software, and cloud integration.
Led cross-functional teams (15+) and $500K–$2M programs, achieving up to 50% cost reduction through in-house system design. Fluent in English, conversational Turkish. Open to relocation.

Professional Experience

Co-Founder & CIO — 2024 – Present

X-Robotiics (IoT workplace intelligence startup, 20+ enterprise deployments) — Istanbul, Turkey

- Architected and launched a production-grade IoT platform deployed across 20+ enterprise environments, defining end-to-end system topology from edge devices through cloud observability with zero critical production failures.
- Owned full technical roadmap: edge computing devices, distributed sensing infrastructure, MQTT communication, telemetry pipelines, and cloud monitoring systems.
- Engineered an offline-first resilience layer enabling uninterrupted edge-device operation and reliable cloud synchronization under unstable network conditions, a key differentiator for enterprise deployments.
- Directed end-to-end product lifecycle: system architecture, manufacturing coordination, calibration workflows, production validation, and enterprise client onboarding.
- Advised enterprise clients on warehouse automation strategy, AMR/AGV fleet sizing, deployment planning, and robotics infrastructure optimization.

Senior Manager, Robotics Infrastructure & Automation — 2021 – 2024

Digikala (Iran's largest e-commerce platform, comparable to Amazon — 600K+ daily parcels) — Tehran, Iran

- Owned architecture and deployment of robotics automation infrastructure across 3 large-scale fulfillment centers, supporting 600K+ daily parcel throughput in 24/7 production environments. Led a cross-functional engineering team of 15+ and managed $500K–$2M hardware/software programs.
- Scaled modular robotic sortation systems from ~2,000 to 6,000+ parcels/hour using distributed PLC control, embedded real-time telemetry pipelines, and sensor fusion, achieving zero unplanned downtime over 18 months.
- Delivered autonomous AMR/AGV platforms integrating SLAM-based navigation, sensor fusion, fleet orchestration, and safety-critical motion control across multi-kilometer fulfillment facilities.
- Improved operational throughput by ~20% and reduced fulfillment errors from ~5% to near-zero via human-in-the-loop automation and closed-loop quality monitoring systems.
- Reduced capital expenditure by ~50% (~$1M+ savings) through modular hardware standardization and in-house actuator development, enabling 3x deployment scale within the same budget.

- Led cross-functional initiatives spanning robotics, embedded systems, backend infrastructure (Python, Linux, Git), mechanical systems, and operations, including technical direction and hiring.

Principal Mechatronics Systems Engineer
2019 – 2021

TARFAND Technical Solutions (biomedical robotics startup, exported to Turkey & Russia)
Tehran, Iran

- Led development and international commercialization of Iran's first domestically produced IVF micromanipulation robotic platform, a system with no domestic precedent and 60% lower cost than comparable European platforms.
- Designed and validated sub-micron motion control systems, achieving 0.2 µm positioning accuracy, a 10x improvement over incumbent platforms using piezoelectric actuation and deterministic RTOS-based control loops.
- Architected closed-loop motion-control systems integrating tremor suppression, strain-gauge feedback, and deterministic embedded control (STM32, CAN Bus, custom PCB design with high-voltage piezo interfaces).
- Commercialized and deployed 38+ biomedical robotic systems across domestic and international markets, validating hardware-software co-design methodology in a regulated clinical environment.

Senior Mechanical Systems Engineer | Industrial Automation & Mobile Systems
2018 – 2019

Barak.co (Large-Scale Bakery Production Systems Manufacturer)
Tehran, Iran

- Designed and deployed industrial automation systems across 500+ large-scale food-processing units using synchronized multi-stage mechanical transmission architectures.
- Led engineering and field deployment of mobile disaster-response kitchen platforms across 60+ emergency-response units, including safety-critical towing, stabilization, and fail-safe emergency braking systems.
- Conducted FMEA-driven reliability engineering and failure analysis across structural, thermal, and mechanical subsystems.

Robotics Systems Architect & Research Assistant
2015 – 2018

Syntech Technology and Innovation Center
Tehran, Iran

- Led robotics architecture and technical mentorship for multidisciplinary robotics teams across RoboCup and applied robotics projects.
- Architected autonomous mobile robotic platforms integrating omnidirectional mobility, robotic manipulation, embedded electronics, and ROS-based autonomy systems.
- Developed STM32-based embedded control and power electronics systems featuring CAN communication, fault protection, and distributed robotics control architectures.
- Engineered robotic manipulator systems with closed-loop motion control, kinematic optimization, and sub-5 mm positioning accuracy.
- Guided cross-functional robotics teams through iterative design reviews, improving engineering quality, manufacturability, and system reliability.

Technical Skills

Robotics & Autonomy: ROS/ROS2 | SLAM | Autonomous Navigation | Fleet Management | Motion Control | Sensor Fusion | AGV/AMR Systems | Robotics Middleware | Gazebo

Mechanical & Simulation: SolidWorks | Abaqus | ANSYS | MSC Adams | GD&T | DFM/DFA

Software & Infrastructure: Python | C/C++ | Linux | Git | MQTT | Telemetry Systems | Distributed Systems | Edge Computing

Embedded & Controls: STM32 | Embedded Linux | RTOS | CAN Bus | PID Control | Motor Drivers | Real-Time Systems | PCB Design | Piezoelectric Actuation

Leadership: Cross-functional team leadership (15+ engineers) | Program budget ownership ($500K–$2M) | Technical roadmap | Hiring | OKR/KPI definition | System architecture | Vendor management

Education

M.Sc. Mechanical Engineering - Applied Design 2019 – 2021
Qazvin Islamic Azad University (QIAU) Qazvin, Iran

- GPA: 3.73/4 | Ranked #1 among all Mechanical Engineering graduate students
- **Thesis:** Design and analysis of a nano-precision fast steering mirror platform using flexure mechanisms and piezoelectric actuation.

B.Sc. Mechanical Engineering - Applied Design 2014 – 2018
Qazvin Islamic Azad University (QIAU) Qazvin, Iran

- Thesis: Design and application of strain wave gearing systems for 6-DoF robotic manipulators.

Languages

Persian
Native

English
TOEFL iBT: 95

Turkish
Conversational

Honors & Recognition

- Ranked 1 among graduate students of Mechanical Engineering, QIAU (2021)
- Member of Iran's National Elites Foundation (INEF)
- First Place – RoboCup @Work League World Competition, **Sydney** (2019)
- First Place – RoboCup Asia Pacific Competition (2018)
- First Place – IranOpen RoboCup Competition (2018)
- Third Place – RoboCup World Competition, **Montreal, Canada** (2018)
- First Place – RoboCup Asia Pacific Competition, **Thailand** (2017)
- First Place – Rescue Robot League World Competition, **China** (2015)